

International Institute of Information Technology, Hyderabad
MA4.101-Real Analysis (Monsoon-2026)

Quiz 1

Time: 45 Minutes

Total Marks: 15

Question 1 [6 Marks]. Prove that $n^2 \leq 2^n$ for every $n \geq 4$. Determine every $n \in \mathbb{N}$ for which it holds.

Solution. Throughout, we apply the induction arguments that begin at n_1 rather than at 0. This is just the ordinary principle of induction applied to the property $P'(k) := P(k + n_1)$.

Auxiliary inequality: $2n + 1 \leq n^2$ for every $n \geq 3$. At $n = 3$ this reads $7 \leq 9$. Assuming it for some $n \geq 3$,

$$2(n + 1) + 1 = (2n + 1) + 2 \leq n^2 + 2 \leq n^2 + 2n + 1 = (n + 1)^2,$$

the last step because $2n + 1 \geq 2$ when $n \geq 3$.

Main inequality. At $n = 4$ we have $4^2 = 16 \leq 16 = 2^4$. Assume $n^2 \leq 2^n$ for some $n \geq 4$. Using the auxiliary inequality, which applies since $n \geq 4 \geq 3$,

$$(n + 1)^2 = n^2 + 2n + 1 \leq n^2 + n^2 = 2n^2 \leq 2 \times 2^n = 2^{n+1}.$$

By induction the inequality holds for all $n \geq 4$.

Failure and full range. At $n = 3$ we have $3^2 = 9 > 8 = 2^3$, so the inequality fails there. The three remaining values are checked directly:

$$n = 0 : 0 \leq 1, \quad n = 1 : 1 \leq 2, \quad n = 2 : 4 \leq 4.$$

Hence the inequality holds exactly for

$$n \in \{0, 1, 2\} \cup \{n : n \geq 4\},$$

that is, for every n except $n = 3$.

Question 2 [9 Marks]. Let $f, g : \mathbb{N} \rightarrow \mathbb{N}$ be functions such that $f(0) = 0$ and

$$f(a) + g(a) + 1 \leq f(a+1) \quad \text{for all } a \in \mathbb{N}, \quad (1)$$

and set $S_{f,g} := \{ f(a) + b : a, b \in \mathbb{N}, b \leq g(a) \}$. In parts (a), (b) and (c) you may use the following two consequences of Eq. (1) *without proof*.

(P1) f is strictly increasing: $f(a) < f(c)$ whenever $a < c$.

(P2) $f(a) + g(a) + 1 \leq f(c)$ whenever $a < c$.

Prove the following.

- (a) [2 Marks] The representation is unique: if $f(a) + b = f(c) + d$ with $b \leq g(a)$ and $d \leq g(c)$, then $a = c$ and $b = d$.
- (b) [4 Marks] If $f(a+1) = f(a) + g(a) + 1$ for every $a \in \mathbb{N}$, then $S_{f,g} = \mathbb{N}$.
- (c) [3 Marks] If $f(a+1) \neq f(a) + g(a) + 1$ for some $a \in \mathbb{N}$, then $S_{f,g} \neq \mathbb{N}$.
- (d) [Bonus, 2 Marks] Prove (P1) and (P2) from Eq. (1). Attempt this only after completing (a), (b) and (c).

Solution.

(a) Suppose $f(a) + b = f(c) + d$ with $b \leq g(a)$ and $d \leq g(c)$, and suppose towards a contradiction that $a \neq c$. By trichotomy, either $a < c$ or $c < a$; since both the hypothesis and the conclusion are unchanged when the pairs (a, b) and (c, d) are exchanged, we may assume $a < c$. Then, using $b \leq g(a)$ and (P2),

$$\begin{aligned} f(a) + b &\leq f(a) + g(a) && [\text{From } b \leq g(a)] \\ &< f(a) + g(a) + 1 && [\text{From } g(a) < g(a) + 1] \\ &\leq f(c) && [\text{From P2 as } a < c] \\ &\leq f(c) + d && [\text{From } 0 \leq d]. \end{aligned}$$

Thus, $f(a) + b < f(c) + d$, contradicting the assumed equality. Hence $a = c$, and then $f(a) + b = f(a) + d$ gives $b = d$ by cancellation of addition in $\mathbb{N}$.

(b) Assume $f(a+1) = f(a) + g(a) + 1$ for every $a \in \mathbb{N}$. Since $S_{f,g} \subseteq \mathbb{N}$ by definition, it suffices to prove $\mathbb{N} \subseteq S_{f,g}$, which we do by induction on $n \in \mathbb{N}$. That is, we will show that $n \in \mathbb{N}$ implies $n \in S_{f,g}$ for all $n \in \mathbb{N}$.

Base case. For $n = 0$: we have $0 \leq g(0)$ as $g(0) \in \mathbb{N}$ and $f(0) = 0$ is given. Then $0 = f(0) + 0 = f(0) + g(0)$. Thus, $0 \in S_{f,g}$.

Inductive step. Now suppose $n \in S_{f,g}$ for some $n \in \mathbb{N}$. Then $n = f(a) + b$ for some $a, b \in \mathbb{N}$ with $b \leq g(a)$. We want to show that $n + 1 \in S_{f,g}$. There are two cases.

- If $b \neq g(a)$, then $b < g(a)$, hence $b + 1 \leq g(a)$, and

$$n + 1 = f(a) + (b + 1) \quad [\text{as } n = f(a) + b \text{ from inductive hypothesis}].$$

Thus, $n + 1 \in S_{f,g}$.

- If $b = g(a)$, then by premise of the question ($f(a) + g(a) + 1 = f(a + 1)$), we have

$$n + 1 = f(a) + g(a) + 1 = f(a + 1) = f(a + 1) + 0,$$

and $0 \leq g(a + 1)$, so again $n + 1 \in S_{f,g}$.

In both cases $n + 1 \in S_{f,g}$, and the induction is complete. Hence $n \in S_{f,g}$ is true for all $n \in \mathbb{N}$.

(c) Here we need to find one natural number n_0 such that $n_0 \notin S_{f,g}$. Fix $a_0 \in \mathbb{N}$ with $f(a_0 + 1) \neq f(a_0) + g(a_0) + 1$ (This exists because of premise of the question). Now from Eq. (1), we have $f(a_0) + g(a_0) + 1 < f(a_0 + 1)$ (strict inequality). Then

$$f(a_0) + g(a_0) + 2 \leq f(a_0 + 1). \quad (2)$$

Now define the natural number $f(a_0) + g(a_0) + 1$ as n_0 . That is

$$n_0 := f(a_0) + g(a_0) + 1.$$

We claim that $n_0 \notin S_{f,g}$. Let $a, b \in \mathbb{N}$ be arbitrary natural numbers such that $b \leq g(a)$. We show $f(a) + b \neq n_0$ for any $a, b \in \mathbb{N}$ with $b \leq g(a)$. This will complete the proof. There can be three cases $a < a_0$, $a = a_0$, and $a > a_0$.

- Suppose $a \leq a_0$. If $a < a_0$, then (P2) gives $f(a) + g(a) + 1 \leq f(a_0)$. Then

$$\begin{aligned} f(a) + g(a) &< f(a_0) & [\text{As } f(a) + g(a) + 1 &\leq f(a_0)] \\ &\leq f(a_0) + g(a_0) & [\text{As } g(a_0) &\geq 0]. \end{aligned}$$

If $a = a_0$ then $f(a) + g(a) = f(a_0) + g(a_0)$. In either case

$$\begin{aligned} f(a) + b &\leq f(a) + g(a) & [\text{As } b &\leq g(a)] \\ &\leq f(a_0) + g(a_0) & [\text{As shown above}] \\ &< n_0 & [\text{As } n_0 = f(a_0) + g(a_0) + 1]. \end{aligned}$$

- Suppose $a > a_0$, i.e. $a \geq a_0 + 1$. By (P1) and Eq. (2),

$$\begin{aligned}
f(a) + b &\geq f(a) && [\text{As } b \geq 0] \\
&\geq f(a_0 + 1) && [\text{Using P1 as } a \geq a_0 + 1] \\
&\geq f(a_0) + g(a_0) + 2 && [\text{Using Eq. (2)}] \\
&\geq n_0 + 1 && [\text{Definition of } n_0] \\
&> n_0.
\end{aligned}$$

So no admissible pair $a, b \in \mathbb{N}$ with $b \leq g(a)$ satisfies $f(a) + b = n_0$, i.e. $n_0 \notin S_{f,g}$. Since $n_0 \in \mathbb{N}$, we conclude $S_{f,g} \neq \mathbb{N}$.

(d) [Bonus]. *Proof of the one-step inequality.* For every $a \in \mathbb{N}$ we have $g(a) + 1 \geq 1$, so by Eq. (1),

$$f(a) < f(a) + g(a) + 1 \leq f(a + 1). \quad (3)$$

In particular $f(a) < f(a + 1)$, and hence $f(a) \leq f(a + 1)$, for every a .

Proof of (P2). Fix a and induct on c , starting at $c = a + 1$. For $c = a + 1$ the claim $f(a) + g(a) + 1 \leq f(a + 1)$ is exactly Eq. (1). Assuming it for some $c > a$, Eq. (3) applied at c gives $f(c) \leq f(c + 1)$, so

$$f(a) + g(a) + 1 \leq f(c) \leq f(c + 1),$$

which is the claim for $c + 1$. By induction (P2) holds for all $c > a$.

Proof of (P1). This now follows from (P2): if $a < c$ then, since $g(a) + 1 \geq 1$,

$$f(a) < f(a) + g(a) + 1 \leq f(c).$$

(Alternatively, (P1) follows directly by inducting on c from Eq. (3) and transitivity.)